

ROGER NG ZI LIANG

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Professional Summary

Engineering graduate with a Bachelor of Engineering (Hons) in Robotics Systems and hands-on experience in mechanical design, systems integration, and process improvement. Familiar with systems engineering principles (INCOSE SEBOK), Lean and root cause analysis, with a strong track record in project documentation, stakeholder coordination, and delivering engineered solutions from concept to fabrication.

Technical Skills

- **CAD & Design:** SolidWorks (Sketching, Assembly, DFM/DFA, GD&T)
- **Prototyping & Testing:** 3D Printing, Validation, Pneumatics (Sysmac Studio)
- **Programming:** Python, C
- **Project Management:** Gantt Charts, BOMs, QA/QC
- **Systems Engineering:** INCOSE SEBOK, systems lifecycle, requirements & interface awareness
- **Process & Systems Engineering:** Lean principles, Six Sigma fundamentals, Root Cause Analysis, process mapping
- **Office Tools:** Microsoft Word, Excel, PowerPoint

Workplace Skills

- Bilingual: Fluent in English and Chinese
- Meticulous, self-motivated, and reliable
- Strong stakeholder communication and presentation skills
- Team-oriented; able to work independently with minimal supervision

Professional Experience

Digital Transformation Analyst

Winner Engineering Pte Ltd – May 2026 - Present

- Designed and implemented AI-assisted solutions and automation workflows utilizing platforms like n8n, applying a Discover-Design-Deploy-Iterate methodology to optimize business processes.
- Managed and monitored self-hosted systems, ensuring operational reliability, cybersecurity, and strict compliance with PDPA and Responsible AI standards.
- Collaborated with internal stakeholders and clients to translate business requirements into user-centric digital applications, driving end-to-end testing, deployment, and troubleshooting.
- Supported organizational adoption by authoring technical documentation, facilitating user training sessions, and mentoring junior project contributors.

Assistant Engineer Intern

WEInnovate Pte Ltd · Jan 2025 – Dec 2025

- Designed 5+ engineered systems and contributed to 10+ cross-disciplinary projects, from concept development to fabrication support
- Applied GD&T and DFM/DFA to ensure manufacturability and design accuracy
- Executed full mechanical assembly and testing for machines/jigs, resolving fit/functional issues via iterative design improvements
- Managed end-to-end procurement and RFQ processes for 100+ components across multiple projects
- Produced and maintained comprehensive engineering project documentation, including BOMs, Gantt charts, Technical drawings, and Machine manuals

- Participated in stakeholder meetings; recorded meeting minutes, aligned action items, and delegated tasks to assigned team members
- Conducted on-site support at customer locations (transporting machines, setup, installation assistance)
- Worked with PLCs, pneumatic systems, sensors, and actuators; conducted basic test using Sysmac Studio
- Supported interdepartmental coordination to streamline project timelines and improve communication flow

Systems Engineer Intern

Destination Services & Intercruises · Apr 2019 – Jul 2019

- Streamlined workflows with Excel VBA and Microsoft Access; reduced lead time by 70%
- Applied root cause analysis to process inefficiencies, contributing to measurable improvements and earning a Certificate of Appreciation
- Developed standardized training documentation to improve onboarding consistency, reducing variability
- Conducted independent R&D on automation tools to support continuous process improvement
- Collaborated with cross-functional teams to improve process reliability and operational stability

Education

Bachelor of Engineering (Hons), Robotics Systems Engineering

Singapore Institute of Technology · Apr 2022 – Apr 2026

- Key Modules: Project Management, AI Robotics, Control Systems
- Tools: SolidWorks, Python, C/C++, MATLAB Simulink, ROS, SEBOK, SysML, Capella

Diploma in Business Process & Systems Engineering

Temasek Polytechnic · Apr 2017 – Apr 2020

- Key Modules: Engineering Analytics, SCM, Marketing Research, Major Project & Internship: Grade A

Technical Projects

- **Robotic Arm Design:** Modeled a robotic arm using SolidWorks; integrated Python-based control
- **Data Mining Project:** Applied RapidMiner to analyze and improve project outcomes
- **Systems Engineering Lifecycle:** Applied SEBOK framework to develop complete systems approach

Additional Experience

Sea Terminal Operator (NSF)

Singapore Armed Forces · Mar 2020 – Mar 2022

- **Certified forklift operator:** directed complex logistics, cranes, and heavy vehicle transport maneuvers under strict safety protocols

Customer Service & Sales (Part-Time)

Various Roles · 2016 – 2020; 2023 – 2025

- Maintained consistent part-time employment across F&B and retail sectors, developing strong communication, situational awareness, and stakeholder interaction skills.